



Improving Course Recommendations: An RBM-Based Explainable Framework for Student Performance in E-Learning

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ABSTRACT

The vast expansion of online learning has increased the need for intelligent systems to guide learners toward suitable learning paths. Although educational recommender systems address this challenge by personalizing recommendations, many existing approaches focus primarily on accuracy and provide limited insight into why recommendations are made. This study proposes EduExplain, an explainable collaborative filtering framework based on a Restricted Boltzmann Machine (RBM) that combines strong predictive performance with neighbor-similarity explanations to generate interpretable recommendations. The framework integrates a prediction engine with an explanation module that produces natural-language justifications derived from the performance of similar learners. We evaluated the framework using a multi-domain experimental design on the OULAD educational dataset and the Goodreads dataset, comparing it against four established baselines. The results show that EduExplain achieves a 20% reduction in prediction error over traditional collaborative filtering (RMSE: 0.873 on OULAD) and delivers high-quality, interpretable explanations (coherence: 0.89), demonstrating that accuracy and explainability can coexist in e-learning systems. Unlike prior approaches that rely on external review data or treat explainability as a post-hoc step, EduExplain embeds explanation generation directly into the RBM pipeline using only interaction data. An ablation study confirmed each component's necessity for both accuracy and interpretability. These findings demonstrate that explainable recommender systems can enhance decision-making in e-learning environments.

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1. INTRODUCTION

In recent years, online education platforms have fundamentally altered the operation of education systems [1]. Students now have the flexibility to access a broader array of educational content and resources that fit their preferences and learning style, anytime and anywhere [2]. However, having so many potentially suitable resources often makes it challenging to deliver effective learning support [3]. To address the challenges learners face in navigating educational resources, recommender systems (RS) have emerged as a key mechanism for personalizing learning. These systems recommend content, courses, or activities tai-

lored to each learner.

Traditional recommender systems, which are commonly used in entertainment or e-commerce, have been adapted for online learning to include CF, item statistical models, content filtering, and hybrid recommender models. More recently, advances in Artificial Intelligence (AI) and Machine Learning (ML) have also incorporated advancements in deep learning (DL), knowledge graphs, and explainable AI (XAI) [4]. These methods are used to enhance the efficacy and transparency of information delivery. Nonetheless, online learning recommender systems encounter several challenges such as the data sparsity issue, the ability to explain recommendations, the cold-start

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problem, the freezing start problem and ethical concerns such as bias and fairness [5, 6].

Despite the widespread use of CF and DL methods in recommender systems, they continue to face technical and educational limitations. The lack of sufficient interaction between users and items for recommendations to be meaningful due to sparse data or the cold-start problem continues to be a technical hurdle decreasing algorithmic trustworthiness and learner satisfaction [7]. Despite their advantages, DL models often operate as “black boxes” providing little insight into their decision-making processes. Users may struggle to trust the reasoning behind recommendations, threatening both ethical design and pedagogical transparency [8]. Most recommender systems prioritize accuracy as the primary evaluation metric, with little focus on educational outcome effectiveness, further complicating the challenge of pedagogical efficiency [9]. While these challenges are all relevant, they are not equally fundamental. Data sparsity and cold-start are largely engineering problems that can be mitigated through better data collection and model design. Explainability, however, is a structural problem. No matter how accurate a model becomes, if learners cannot understand why a course is recommended, they cannot meaningfully trust or act on that recommendation. This paper therefore targets the accuracy–explainability trade-off as the central research gap.

The core challenge in educational recommender systems is the accuracy-explainability trade-off. On one side, deep learning models such as neural collaborative filtering and autoencoders have achieved strong predictive performance. Still, they operate as black boxes, providing no insight into why a specific course is suggested to a particular learner [10, 11]. On the other side, explainable models that generate interpretable justifications typically sacrifice predictive power, or depend on external data sources such as user reviews, which are simply not available in most learning management systems [12, 13]. The result is a field where systems are either accurate but unclear, or interpretable but weak. Neither is acceptable in an educational context, where a learner’s decision to enroll in a course may have real consequences for their academic trajectory. Bridging this gap, without sacrificing either accuracy or interpretability, and without requiring external data, is the central problem this paper addresses.

This raises a precise research question: *Can a single recommender system achieve state-of-the-art predictive accuracy and generate meaningful, personalized explanations using only student interaction data, with no dependency on external review text?* This paper proposes an effective explainable collaborative filtering model, built on the RBM algorithm, called EduExplain, for e-learning recommendation platforms. The framework enables learners to under-

stand not only the recommendations they receive but also the reasoning behind them, thereby enhancing trust and pedagogical transparency. The proposed model was evaluated and compared with other existing similar research work addressing the same problem and using the same dataset or algorithm. We show that the proposed model outperforms existing baselines in predictive accuracy. The major contributions of this work are summarized below.

1. We propose EduExplain, a framework integrating RBM with neighbor-based explanations for high predictive accuracy (20% improvement over traditional CF) and interpretable justifications without requiring external sources like reviews.
2. We introduce a dual-threshold mechanism ensuring prediction confidence (≥ 0.7) and explainability (≥ 0.65), improving precision by $\sim 4.8\%$ over single-threshold methods.
3. We conduct a multi-dimensional evaluation on the OULAD and Goodreads datasets, including explanation metrics and ablation analysis.

Unique Contribution and Significance: This work presents an integration of RBM-based collaborative filtering with neighbor-similarity explanations for educational recommender systems. It achieves predictive accuracy without external data sources like reviews, while offering interpretable natural-language justifications linked to pedagogical outcomes.

The rest of this paper is organized as follows. Section 2 reviews the related work. Section 3 details the methodology. Section 4 describes the experimental setup and datasets. Section 5 presents results and discussion. Section 6 presents the conclusion, limitations, and future research.

2. LITERATURE REVIEW

We begin by reviewing traditional and deep-learning collaborative filtering methods, which form our work’s predictive foundation. Accurately predicting student performance is a primary goal for providing timely support in online E-learning [14]. To achieve this, educational systems must analyze the large volumes of data generated by students to suggest helpful materials [15, 16]. Several studies have attempted to address portions of the problem. Within online learning, it is crucial to use adaptive systems that recommend material based on a student’s prior knowledge [17]. Other researchers have developed systems to align material recommendations with a student’s knowledge structure [18]. Therefore, future systems must achieve both high predictive performance and interpretability to foster user trust.

2.1 Traditional and Deep Learning Collaborative Filtering

Deep learning is a technology for discovering hidden characteristics in data, used for dimensionality reduction and feature extraction. It has lately been

Table 1: List of references for past studies.

(Year), Ref.	Methodology	Application Domain	Multi-Domain	Limitation	Explainable
Edu Explain (2026)	RBM + CF	E-Learning	YES	Requires 100-200 interactions/user; Retraining for catalog changes	YES
(2025) [19]	Deep FM + SVD++ Hybrid	Multi-domain (Tourism, Commerce, Edu)	YES	Multi-criteria data needed; Complex tuning	PARTIAL (FM only)
(2024) [18]	LSTM + aSDAE + Dual-Matrix + BPNN	E-Learning (OULAD)	No	Complex 3-stage; High cost; Platform-specific	NO
(2024) [3]	BiLSTM + GRU + Attention + Compression	E-Learning (OULAD)	No	Manual separation; High memory; Difficult tuning	NO
(2024) [20]	Hybrid Filtering Approach: CF+CBF Attention mechanism	E-Learning (Educational)	YES	Limited details; Unclear cold-start	PARTIAL (Features)
(2023) [21]	KNN, SVD, Neural CF, Matrix Factorization with Gradient Descent;	E-Learning (Coursera)	NO	Cold-start not addressed; Single platform	NO
(2023) [17]	Graph CNN + Node Embedding	E-Learning (MOOCs)	YES	Complex graph $O(n^2)$. Focuses purely on prediction accuracy without explanation. black-box	NO
(2023) [22]	Transformer + Retrieval + Aspect	Traveling (Amazon, Yelp, TripAdvisor)	No	Requires large text data; High overhead	YES
(2020) [5]	Deep NN + Embedding + MLP	E-Learning (Books)	No	Cold-start, Black-box deep model; relies only on user-item ratings	NO
(2020) [8]	LSTM + MNN + MF + Reliability	Movies (Netflix) (MovieLens) (FilmTrust)	No	Entertainment-focused; 3-stage complexity	NO
(2019) [15]	VAE + Neural CF	Traditional Classroom	No	small-scale dataset Single-domain (Education) No generalization	NO

used intensively in the field of recommender systems [12]. CF forms the basis of many recommender systems; classical CF has problems such as data sparsity and cold-start. These issues are common in educational settings [19]. Matrix factorization and latent factor models improved prediction performance by capturing latent relationships in how students and tasks relate [20, 21]. Researchers use deep learning to fix these limits, often matching or surpassing traditional machine learning models [22]. newer studies mix Collaborative Filtering (CF) with deep learning (like neural CF or autoencoders). Moreover, they use hybrid models integrating (CF with Naive Bayes, knowledge graphs and rule-based systems). These changes help fix problems such as missing data or new tasks, and improve predictions more accurate [15] [18]. since RBM beat older methods like Singular Value Decomposition (SVD) for rating prediction. Cheng et al. showed that combining CF with RBMs improves accuracy [23]. Mawane et al. proposed an unsupervised deep CF model to improve suggestions for e-learning students [24]. These studies boost prediction but act like a “black box.” They do not ex-

plain why a course is suggested. At the end, this lack of explanation hurts trust in education. [16] perform deep feed-forward networks and LSTM models to predict academic performance from Open University clickstream logs, reporting accuracies up to about 93% and clear gains over traditional classifiers. While collaborative filtering techniques improve predictive accuracy, they often lack interpretability, which motivates the development of explainable recommender systems reviewed next. Overall, these methods fail to meet educational needs for two core reasons. First, deep models such as autoencoders and neural CF optimize for rating reconstruction but provide no mechanism for justifying recommendations in pedagogically meaningful terms. Second, because they treat the user as a passive recipient of scores rather than a learner who needs to understand why a resource is suggested, they structurally cannot address the trust barrier inherent in educational decision-making.

2.2 Explainable Recommender Systems and Multi-Level Evaluation

The push toward transparent AI has made explainable recommender systems (XRS) a central research direction. A systematic by Zhang and Chen g [25] organizes the field along explanation style, generation strategy, and evaluation criteria, and documents gains in user satisfaction, trust, and decision quality when explanations are well designed. Early work integrated explanations directly into the predictor, with the Explainable RBM of Abdollahi and Nasraoui [26] deriving justifications from latent representations rather than from a separate post-hoc analyzer. Recent state-of-the-art systems extend this line in three architectural directions, each representative of a broader family XRec [27] uses a large language model to generate fluent natural-language justifications, but it requires user-generated review text at inference and access to a capable LLM. Neither resource is reliably available in learning management systems, where students rarely write course reviews and where institutional constraints on latency, per-query cost, vendor lock-in, and student-data privacy under FERPA and GDPR limit the use of commercial LLM pipelines. O3ERS [13] addresses explainability in online learning, yet its target is generic item recommendation rather than student-performance prediction; its explanation channel is not tied to a pedagogical outcome variable, which is the variable that matters in educational deployment. Knowledge-graph and LSTM hybrids exemplified by [10] anchor explanations in graph relations, but they presuppose a curated educational knowledge graph that is rarely maintained at the course level and is costly to construct. These systems are representative of language-based, online-learning, and graph-based architectures, and share a common property: explanation is grafted onto a predictor and depends on a resource beyond the interaction matrix. EduExplain takes a different position on each axis: it uses only interaction data, generates explanations from the same RBM latent space that produces predictions, and is evaluated against a pedagogical outcome variable.

A second concern in evaluating explainable recommenders is the choice of ranking cut-off. Prior work commonly reports performance at small list sizes such as Top@10 or Top@20 [28], [29], although deployment scenarios call for varied list sizes and stability across cut-offs remains an open question [30]. We therefore evaluate Top@10 across two distinct domains and examine how the metrics behave at higher cut-offs.

In summary, existing explainable recommender systems fail in educational contexts for three identifiable reasons: (1) they depend on user-generated review text that is unavailable in learning management systems; (2) their explanation mechanisms are post-hoc and disconnected from the underlying prediction process, reducing consistency; and (3) they are

not evaluated against pedagogical outcomes, making their educational value unverifiable. These structural gaps directly motivate the design of EduExplain.

2.3 Synthesis and Our Contribution

The literature above develops along two threads. The first refines deep collaborative filtering: RBM-based models in particular have been shown to surpass classical matrix factorization for rating prediction [23], [24], but they remain opaque about the reasoning behind any individual recommendation. The second develops explainable recommenders whose justifications are linguistically rich [27] or graph-grounded [10], but rely on resources educational platforms do not typically maintain, or target a recommendation task other than student-performance prediction [13]. Neither thread, taken alone, meets the dual requirement of an educational recommender: predictive accuracy on sparse interaction data, and pedagogically meaningful justifications that students and instructors can verify against learning outcomes.

EduExplain sits between these threads. It uses an RBM to model student-item interactions. It derives explanations from the same latent representations used for prediction, with peer similarity in that latent space supplying the substantive content of each justification. The framework requires only interaction data, places explanation in the same computation that produces the prediction rather than as a separate post-hoc layer, and is evaluated against pedagogical outcomes alongside standard ranking metrics. Table 1 situates this design against the prior studies surveyed above and motivates the methodology developed in section 3.

3. METHODOLOGY

3.1 Overview of EduExplain

EduExplain is a novel explainable recommendation framework that integrates the strong predictive power of RBM with an explainable model to support student performance. This model creates an explanation for every given recommendation. As a result, students can understand why they received a particular recommendation. This clarity enhances the learning environment and supports more informed decision-making. The architecture of the EduExplain framework is designed to join two essential capabilities: the predictive accuracy of RBMs and a neighbor-based explanation mechanism. The system operates through four sequential phases, as illustrated in Fig. 1. These phases form a unified pipeline that transforms raw student interactions into accurate and interpretable recommendations. Fig. 1. provides an overview of the complete EduExplain pipeline, illustrating the flow from raw data preprocessing to the generation of explained recommendations.

1. Data Fusion and Preprocessing, in this phase, raw

data from two datasets are cleaned, normalized, and merged into a unified interaction matrix.

2. **RBM Prediction Engine**, in this phase, the unified data is processed by a Restricted Boltzmann Machine, which learns latent patterns in user-item interactions and generates initial probability scores for unseen items.
3. **Explanation Generation**, which creates transparency. It identifies a student's nearest neighbors, calculates the explanation score based on their preferences, and then converts this score into a natural language justification.
4. **Dual-Threshold Filtering**, in this phase, a dual-threshold filter is applied to candidate items from the RBM engine, using both the prediction score and the explanation score to select the final high-quality recommendations.

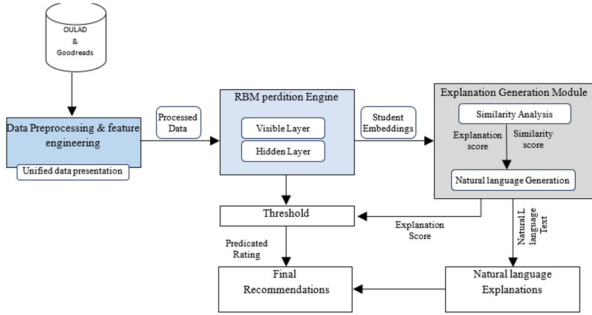


Fig.1: The EduExplain framework.

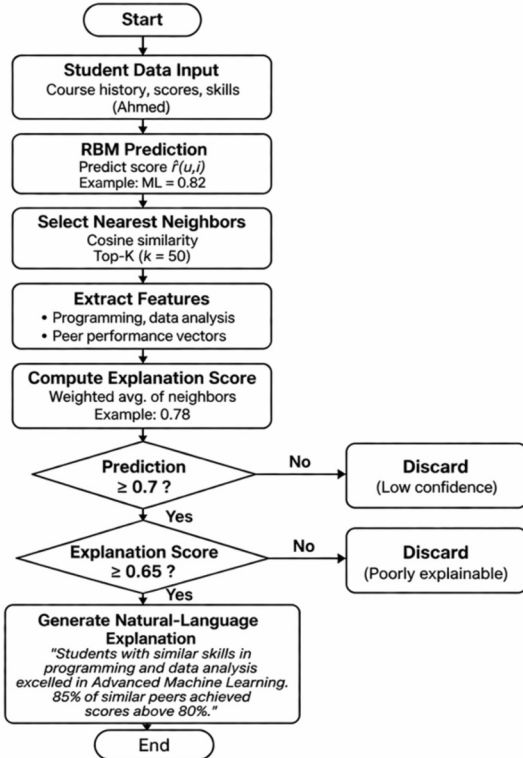


Fig.2: Detailed flowchart of the EduExplain recommendation pipeline.

The diagram Fig.2. illustrates the full process from student data input through RBM prediction, nearest-neighbour selection, explanation-score computation, dual-threshold filtering (prediction ≥ 0.7 and explanation score ≥ 0.65), and generation of the natural-language explanation. Concrete values from the illustrative case study (Sara) are included for clarity.

3.2 Data Preprocessing

Data preprocessing is an essential step for developing an effective predictive model. First, for dataset consistency, samples with incorrect date fields were removed. Both datasets (OULAD and Goodreads), were initially standardized and cleaned. For OULAD, all student-related tables (studentInfo, studentVLE, studentAssessment, and courses) were merged using `id_student` and `code_module`. missing or invalid entries were removed; categorical variables such as module code and gender were label-encoded. For the Goodreads dataset, the interaction file (goodreads_interactions.csv) was merged with the metadata file (books_metadata.csv) using `book_id`. the *duplicate* interaction values and books without ratings were excluded. In addition, to reduce the dimensionality and complexity of the model, integer encoding was used instead of the classification data in the demographic data.

3.3 Feature selection and normalization

In the feature selection and normalization process, key behavioral indicators were chosen from the OULAD dataset, including the number of VLE interactions, assessment completion score, and final result label. For the Goodreads dataset, selected features comprised user-book rating, number of text reviews, and average book rating. All numerical features from both datasets were normalized to a $[0, 1]$ range using min-max scaling to ensure balanced learning across domains as given by Equation (1).

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

3.4 Unified matrix representation, training procedure and sparsity strategy

Both datasets are typically transformed into a Unified Interaction Matrix with the structure (user_id, item_id, rating), which allows collaborative filtering models to process data from multiple sources in a consistent format; this is captured in Equation (2). To keep the distinction between domains, an additional domain indicator feature is often added (e.g., domain = 0 for OULAD, 1 for Goodreads). This practice enables models such as RBM to learn shared latent structures across domains while retaining awareness of the data origin, improving both recommendation accuracy and the ability to transfer knowl-

edge between domains. To address sparsity and domain imbalance, we used stratified downsampling to match interaction scales. A balanced subset included 3,000 OULAD learners across 30 courses and 10,000 Goodreads users across 1,000 books. This balance reduced bias toward Goodreads, stabilized training, and lowered computation while keeping representative interaction patterns. To further enhance model robustness, we applied data augmentation techniques such as mild noise injection to the interaction matrix, which improved generalization, particularly for sparse user profiles.

$$R = \begin{cases} R^{edu} & \text{for educational domain} \\ R^{books} & \text{for literary domain} \end{cases} \quad (2)$$

3.5 Model Architecture

3.5.1 RBM Prediction model

The core of the prediction process in the Edu-Explain framework is the (RBM), a two-layer, undirected probabilistic graphical model, which is responsible for learning the hidden pattern between students and items. Our prediction method builds upon the RBM model for collaborative filtering proposed by [23]. The model consists of a visible layer, v , which represents the user's categorical rating values for all items (courses/books); the hidden layer h , captures the latent features influencing user preferences. Equation (3) defines the energy function for the joint configuration (v, h) .

$$E(v, h) = -\sum_i a_i v_i - \sum_j b_j h_j - \sum_i \sum_j v_i w_{ij} h_j \quad (3)$$

Then, the conditional activation probabilities are given by Equation (4).

$$\begin{aligned} P(h_j = 1|v) &= \sigma(b_j + \sum_i w_{ij} v_i), \\ P(v_i = 1|h) &= \sigma(a_i + \sum_j w_{ij} h_j) \end{aligned} \quad (4)$$

where the sigmoid function or logistic function is defined as $\sigma(x) = 1/(1+e^{-x})$ subsequently, the model reconstructs the visible units, including the missing entries. The resulting probabilities for the unrated items constitute the final predictions. The model is trained using the Contrastive Divergence algorithm (CD-1) [24], which approximates the gradient of the likelihood by measuring the reconstruction error between the input data and its reconstruction. This approach enables the RBM to efficiently learn the joint distribution of visible and hidden units. Training seeks to maximize the likelihood of observed user interactions by minimizing the reconstruction error. Equation (5) formalizes how the RBM updates its energy parameters during learning. The RBM is optimized using mini-batch gradient descent with a learning rate of 0.01 over 50 epochs. A batch size of 64

is employed during training to stabilize gradient updates.

$$\Delta w_{ij} = \epsilon (\langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{recon}}) \quad (5)$$

where ϵ is the learning rate, after training, the RBM produces two key outputs: 1. probabilistic predicted ratings for items that students have not rated, and 2- latent representations (embeddings) for users and items. These embeddings are later used by the explanation module to identify similar learners.

3.5.2 Explanation module development

We created an explanation component to address the black-box aspect of RBMs, drawing inspiration from the explainable AI tenets presented in Equation (6). We create explanations for every suggested course c to target students using the following procedure. For neighborhood formation, we identify the k -nearest neighbors $N_u(s)$ of student s using cosine similarity on their rating vectors, following the collaborative filtering paradigm established by [11].

$$\text{sim}(u, v) = \frac{r_u \cdot r_v}{\|r_u\| \|r_v\|} \quad (6)$$

Then we compute the explanation score for a recommended course c as indicated in Equation (7).

$$\text{ExplanationScore}(u, i) = \frac{\sum_{v \in N_k(u)} \text{sim}(u, v) \cdot r_{v,i}}{\sum_{v \in N_k(u)} |\text{sim}(u, v)|} \quad (7)$$

Where $\text{Sim}(s, u)$ represents the cosine similarity between students s and u , and $r_{u,c}$ is the rating of user u for course c . Finally, following the trend of user-friendly explanations highlighted in [5], we convert the explanation score into natural language templates.

To generate relevant explanations that support student decision-making, there exist several factors affecting the quality of generation. The explanation generation pipeline (Extract features $\rightarrow$ Populate templates $\rightarrow$ Apply thresholds $\rightarrow$ Generate final explanation) in our framework consists of four different steps. These factors and steps are shown in Fig. 3.

3.5.3 Dual threshold filtering approach

We employ a multi-stage thresholding approach to ensure accurate recommendations and understandable explanation. The threshold values were empirically determined through a grid search to balance precision and recall. First, the approach selects items whose predicted rating is $\theta_{prea} \geq 0.7$, as indicated in Equation (8), and the explanation score is $\theta_{exp} \geq 0.65$, as indicated in Equation (9).

$$C_{\text{candidates}} = \{i \in I | \hat{r}_{u,i} \geq \theta_{pred}\} \quad (8)$$

Variable calculation <i>OULAD</i> = {skills, course, percent, threshold, count, and related courses} <i>Goodreads</i> = {similar books, book, percent, genres and author style}
Initial explanation example: <i>- Educational: Students with similar skills in {skills} excelled in {course}. {percent}% of similar peers achieved scores above {threshold}%.</i> <i>- Literary: Readers who enjoyed {similar_books} rated {book} highly. {percent}% of similar readers gave it 4+ stars.</i>
Threshold filtering <i>Prediction Threshold:</i> $\theta_{pred} \geq 0.7$ {high recommended}, $0.65 \leq \theta_{pred} < 0.7$ {medium recommended}, $\theta_{pred} < 0.65$ {Low recommended}. <i>Explanation Threshold:</i> $\theta_{exp} \geq 0.65$ Explanation withheld: $\theta_{exp} < 0.65$ Item receives explanation only if: $\theta_{pred} \geq 0.65) \wedge (\theta_{exp} \geq 0.65)$
Final explanation real example: <i>Students with similar skills in programming and data analysis excelled in Advanced Machine Learning. 85% of similar peers achieved scores above 80%.</i> <i>Based on your interest in philosophical science fiction, 'Solaris' by Stanisław Lem comes highly recommended. 81% of readers who enjoy contemplative sci-fi rate it 4+ stars, and it shares thematic elements with works you've previously rated highly.</i>

Fig.3: The Explanation generation.

$$C_{final} = \{i \in C_{candidates} \mid \text{ExplanationScore}(u, i) \geq \theta_{exp}\} \quad (9)$$

where $\theta_{pred} = 0.7$ and $\theta_{exp} = 0.65$ are prediction and explanation thresholds, respectively. The dual-threshold filtering process is visually illustrated in Fig. 2.

3.6 Illustrative Case Study:

Consider a student “Sara” who has completed three OULAD modules with high engagement in programming and statistics. EduExplain processes her as follows:

- RBM Prediction:** Sara’s interaction vector is fed into the RBM. The hidden layer activates latent features associated with quantitative reasoning. The model predicts a high score (0.82) for the course (Advanced Machine Learning).
- Neighbor Identification:** Cosine similarity identifies $k = 30$ peers whose interaction patterns most resemble Sara’s. Of these, 85% completed (Advanced Machine Learning) with scores above 80%.

- Threshold Filtering:** The predicted score ($0.82 \geq \theta_{pred} = 0.70$) and explanation score ($0.79 \geq \theta_{exp} = 0.65$) both pass the dual threshold.
- Natural-Language Output:** Students with similar skills in programming and data analysis excelled in Advanced Machine Learning. 85% of similar peers achieved scores above 80%.
- This case illustrates how each pipeline stage contributes to a recommendation that is both accurate and transparent.

The complete flow for Sara’s recommendation, including all thresholds and the final natural-language explanation, is shown step-by-step in Fig. 2.

4. EXPERIMENTAL SETUP

4.1 Dataset description

This study utilizes two datasets from distinct domains to construct a multi-domain recommendation model. The first is the (OULAD), a comprehensive resource containing demographic data (e.g., age, gender), course registrations, assessment scores, and over 10.6 million virtual learning environments (VLE). It has interaction logs for 32,593 students across 22 courses, with over 10.6 million records of student interactions. The second dataset is derived from Goodreads, providing a rich repository of book metadata (including titles, genres, and publication years, in addition to user-generated content such as ratings, reviews, and likes. With information on over 2.3 million books and 876,000 users, the Goodreads dataset is helpful for modeling literary popularity and user. The experiments utilized the final unified and sampled dataset, which contains interactions from 3,000 OULAD learners and 10,000 Goodreads users. The unified dataset was split into an 80% training set for model learning and a 20% testing set for evaluation. The split was performed using stratified sampling to preserve the user-item distributions.

OULAD was selected for its rich student behavioral data, including VLE interactions, assessment scores, and demographics, making it directly relevant to educational recommender evaluation. Goodreads was selected to test cross-domain generalizability, as its user-book rating structure mirrors student-course interaction data. Together, they provide both data-sparse and data-dense conditions that reflect realistic deployment scenarios.

4.2 Implementation Setup

EduExplain was developed using PyTorch 1.9 environment, the NLTK and spaCy libraries were used for natural-language explanation module. All experiments were conducted on a machine equipped with an Intel i9 CPU and an NVIDIA RTX 3080 GPU to facilitate efficient training. The hyperparameters for the RBM module, explanation module including pre-

diction threshold, explanation threshold and neighborhood size were tuned through grid search, maximizing the F1-score on the validation set. The final values used were: Hidden units: 500, Learning rate: 0.01, Batch size: 64, Epochs: 50, and Contrastive Divergence steps (CD-k): 1, with Neighborhood size (k): 50. While Stochastic Gradient Descent (SGD) is the conventional choice for training RBMs with Contrastive Divergence, we found that using the Adam optimizer led to more stable convergence and slightly better final performance in our experiments. To ensure reproducibility and fair comparison, all baseline models were trained and evaluated under the same background and conditions. To facilitate efficient model training and evaluation, all experiments are conducted using the same workstation, with the following hardware configuration as shown in Table 2 and hyperparameter setting as shown in Table 3. The sensitivity of results to random initialization is reported in Table 10, and the sensitivity to neighborhood size k is reported in Table 11 and discussed further in section 5.5.

Table 2: Hardware configuration.

Component	Specification
CPU	Intel Core i7 (8 cores, 3.2 GHz)
GPU	NVIDIA RTX 3060 (12 GB VRAM)
RAM	32 GB
Storage	100 GB SSD
OS	Ubuntu 22.04 LTS
Frameworks	Python 3.9, PyTorch, scikit-learn, Surprise, spaCy

Table 3: Hyperparameter Settings for Our RBM Model.

Hyperparameter	Value	Description
Hidden units (D)	500	Number of latent features
Learning rate	0.01	Adam optimizer setting
Batch size	64	Mini-batch training
Epochs	50	Training iterations
K(CD-k)	1	Contrastive divergence steps
Neighborhood size	50	Similar students for explanations

4.3 Baseline models

To evaluate the performance of EduExplain model, we conducted a comparative analysis against four baseline models.

1. Standard RBM [12], uses a RBM to learn a probabilistic model that can predict missing user-item interactions.
2. User-based CF [25], predicts a user’s preference for an item based on the ratings of similar users.
3. SVD [26], maps both users and items into a shared latent factor space, where their interaction is modeled as an inner product.
4. AutoRec [27], is an autoencoder-based model, that reconstructs user ratings from partially observed vectors using non-linear transformations.

To enable a controlled comparison of latent representations under a common explanation interface, the four baselines are also evaluated using a shared neighbor-based explainer that uses the same neighborhood formation (Eq. 6), explanation scoring (Eq. 7), and template procedure as EduExplain. This shared-explainer setup is used solely in the explanation-quality and is not applied in the prediction or ranking comparisons of Sections 5.1, 5.2, and 5.3.

4.4 Evaluation Metrics

We perform two separate evaluations to ensure fair comparison. Predictive performance is evaluated on the original baseline models without explanation components. Explanation quality is evaluated by applying the same neighbor-based explanation generator (Eq. 6 and Eq.7, and template-based generation) to all models. This separation prevents the explanation module from unfairly influencing the predictive comparison and allows clear assessment of each model’s intrinsic predictive capability versus its explainability when equipped with a comparable explanation mechanism. baseline models were implemented using the same preprocessed data to ensure a consistent comparison with EduExplain. Model performance was assessed across multiple dimensions: Predictive accuracy: RMSE and MAE, ranking performance: Precision@10, Recall@10, and Normalized Discounted Cumulative Gain (NDCG@10) and Diversity and coverage: Coverage@100 and Novelty@100. To assess the semantic alignment between the generated explanations and the underlying recommendation rationale, we used measuring explanation coherence. This metric evaluates how well the explanation reflects both the recommended items content and the users academic profile. Coherence is computed by embedding the explanation text and reference documents. The cosine similarity between these embeddings is used as a coherence score. Higher values means that the explanation semantically aligns with the student’s history and the recommended course, suggesting a more faithful and informative justification.

4.4.1 Explanation Quality Evaluation Framework

To evaluate the quality of generated explanations, we employ four objective metrics capturing complementary dimensions of explainability in our hybrid recommendation system. First, explanation coherence measures semantic alignment between explanations and users’ hybrid profiles, which integrate OULAD learning patterns with Goodreads literary preferences via cosine similarity of their sentence embeddings. Second, explanation consistency assesses

the stability of explanations by clustering users with similar hybrid profiles and calculating the average pairwise similarity of explanations within each cluster. Third, explanation diversity evaluates the several justification styles by measuring the ratio of unique explanation forms to total explanations per user. Fourth, explanation personalization evaluates tailoring to individual users by calculating the cosine distance between a user's average explanation embedding and the global average across all users. For computation, we generate textual explanations using templates populated with hybrid features, embed all texts using Sentence-BERT, and Subsequently calculate similarity and diversity measures.

5. RESULTS AND DISCUSSION

Section 5 reports two families of comparisons. Sections 5.1 and 5.2 compare the original baselines, in their published form and without any imposed components, on predictive accuracy, ranking, coverage, and novelty. Section 5.3 evaluates explanation quality using a controlled comparison in which the four baselines are equipped with a shared neighbor-based explainer to isolate the contribution of the latent representations.

Based on the results of the comparison, an evaluation was conducted using root mean square error. The detail of results and discussion are presented in this section, comprehensive evaluation of the proposed EduExplain framework on the OULAD and Goodreads datasets. We compare the EduExplain model performance with several baseline CF models across different dimensions including (predictive accuracy, explanation quality, coverage, novelty, and computational efficiency). using standard metrics, including (RMSE, MAE, Precision@10, Recall@10, and NDCG@10).

5.1 Predictive Performance Analysis

Tables 4 and 5 report each baseline (User-based CF, MF-SVD, AutoRec, Standard RBM) in its originally published form, with no auxiliary explanation module attached; the imposed explainer used in the explanation-quality comparison of section 5.3 plays no role in these tables. The proposed EduExplain Model excelled in performance metrics as compared to the baseline models. Table 4 demonstrates the superior predictive accuracy of EduExplain. it achieves the lowest error rates for RMSE (0.873 for OULAD) and (0.815 for Goodreads). The analysis findings show a notable improvement of 20.3% and 24.3%, respectively, compared to the traditional CF.

The EduExplain exceeds the explanation of the normal RBM, where the RBM ignores explanation components; in our approach, with the integration of both models, we aim to enhance the predictive accuracy. Thus, the enhanced accuracy would directly

contribute to the increase in competitive ranking performance. Therefore, EduExplain is expected to attain the highest scores across all ranking metrics (Precision@10, Recall@10 and NDCG@10). As shown in Table 5, three tests were used to compare between the five methods. The values in the table indicates clear gaps among the five methods. Method User-based CF scored 0.32 on Precision@10, 0.045 on Recall@10, and 0.35 on NDCG@10, while, method such as Matrix Factorization (SVD) reach to 0.37, 0.058, and 0.41. The rise is small, but it shows some gain over User-based CF. the real change appears with method AutoRec. Its scores rose to 0.42, 0.067, and 0.46. This marks a steady rise of more than 25 percent over User-based CF in Precision@10 and NDCG@10. The Standard RBM stayed close to AutoRec, with 0.41, 0.066, and 0.45. The small decrease shows that both methods work at a similar level. EduExplain (ours) stood out. It scored 0.44, 0.071, and 0.47. These are the highest values in all tests. The gains over User-based CF reach more than 30 percent. also passed AutoRec by about 5 percent in Precision@10. This value shows that EduExplain is the strongest and most steady ranking.

RBM outperforms traditional CF mainly because it learns robust latent representations that capture complex user-item interactions and handles data sparsity effectively through probabilistic modeling. This strength in handling sparsity becomes especially clear when examining performance across different user activity levels. Performance on User Groups:

- Cold-start users (<5 interactions): RMSE = 0.91, Precision@10 = 0.38
- Active users (≥ 50 interactions): RMSE = 0.85, Precision@10 = 0.47

These results show reasonable performance even with sparse data, though accuracy improves with more interactions.

We report per-group metrics. EduExplain shows larger relative gains for moderate and active users, where sufficient interaction history allows the RBM model to capture richer latent patterns and enables more reliable neighbor selection. For cold-start users, the improvements are smaller but still observable, as the neighbor-based explanation mechanism can leverage limited interaction signals to propagate preferences from similar users.

The superior performance of EduExplain over traditional CF and even Standard RBM can be explained by two mechanisms. First, the RBM learns robust latent representations that capture complex, non-linear user-item interaction patterns that simpler models such as SVD cannot model. Second, the integration of the explanation module introduces an additional regularization effect: by requiring recommendations to satisfy both a prediction threshold and an explanation threshold, the dual-threshold filter implicitly removes low-confidence predictions that

would otherwise inflate error rates. This is consistent with XAI theory, which holds that constraining a model to produce interpretable outputs forces it to rely on more generalizable, stable features rather than spurious correlations.

Table 4: summarizes the predictive accuracy comparison.

Method	RMSE		MAE	
	OULAD	Goodreads	OULAD	Goodreads
User-based CF	1.095	1.078	0.872	0.856
MF (SVD)	0.972	0.941	0.784	0.753
AutoRec	0.897	0.861	0.716	0.687
Standard RBM	0.902	0.868	0.722	0.692
EduExplain (ours)	0.873	0.815	0.698	0.656

Table 5: Ranking Performance Comparison.

Method	Precision @10	Recall @10	NDCG @10
User-based CF	0.32	0.045	0.35
MF (SVD)	0.37	0.058	0.41
AutoRec	0.42	0.067	0.46
Standard RBM	0.41	0.066	0.45
EduExplain (ours)	0.44	0.071	0.47

5.2 Coverage and Novelty Analysis

To fairly assess, we used two metrics (coverage and novelty) to see how well EduExplain supports diverse recommendations. Table 6 summarizes the Coverage and Novelty for each model compares our model. EduExplain reached 57% coverage, a 16.29% average improvement. This means the system recommends a wider set of items overall. On the sparse OULAD dataset, the coverage gap is clearer: EduExplain outperformed the closest baseline, AutoRec, by 3 percentage points. For Novelty analysis, EduExplain excels in novelty metrics, reaching 8.9, which are higher than all baseline models. This result shows that our model is effective in mitigating the bias by recommending more items that are not mainstream. Notably, on the data-sparse datasets, the novelty score of EduExplain is 12.93 percentage higher than that of the baseline on average, indicating its ability to recommend less familiar but potentially valuable courses. This suggests the system can expose students to learning opportunities they might not encounter.

Table 6: Recommendation Diversity Metrics Comparison.

Method	Coverage@100	Novelty@100
User-based CF	0.42	6.8
MF (SVD)	0.49	7.9
AutoRec	0.54	8.6
Standard RBM	0.53	8.5
EduExplain (ours)	0.57	8.9

5.3 Explanation Quality

Table 7 explains the effectiveness and usefulness of EduExplain model for end users. To enable a fair explainability comparison, the same neighbor-based explanation generator (based on Eq. 6 and 7 together with template-based natural language generation) was applied uniformly to all baseline models as a post-hoc module.

The model assessment has been carried out by using the four objectives’ metrics mentioned in subsection 4.4.1. as a criterion. According to results shown in Table 7, the model proposed in this paper performs strongly across all dimensions. First, with regards to coherence, EduExplain scores 0.89, which indicates an improvement of approximately 30.9% over the lowest baseline User-based CF and 21% points over the strongest baseline AutoRec. these results signify that the explanations better match the content of the recommended items. Second, regarding the consistency, the model achieves 0.85%, exceeding AutoRec’s score of 0.65%. The increase indicates the enhancement in the stability across similar user-item. Third, in terms of diversity EduExplain reaches 0.82%, exceeding the score of AutoRec 0.62%, confirming the strong engagement. fourth, for the personalization, scores 0.86, representing a 34.4% exceeding the average baseline. This result suggests the model’s capacity to generate user-specific explanations, and not relying on generic templates. Thus, EduExplain outperforms the baselines across all evaluation dimensions.

A key theoretical reason for this advantage lies in the distinction between intrinsic and post-hoc explainability. In EduExplain, explanations are generated directly from the latent representations learned by the RBM during training, meaning the explanation mechanism is coupled with the prediction process itself. In contrast, when the same explanation generator is applied to baseline models such as SVD or AutoRec, it operates as a post-hoc layer on top of representations that were never optimized for explainability. This structural difference explains why EduExplain achieves coherence of 0.89 compared to 0.68 for AutoRec, even when both use the same explanation template: the underlying representations that feed the explanation module are simply richer and more semantically aligned in EduExplain. This finding supports the broader XAI principle that ex-

planation quality is not separable from model architecture, and that explainability must be designed into the system.

EduExplain follows the intrinsic explainability paradigm by coupling representation learning in the RBM with neighbor-based explanation generation. This integration encourages the model to learn latent representations that are not only effective for prediction but also semantically meaningful for identifying pedagogically similar peers. Consequently, the resulting explanations are more faithful and consistent compared to post-hoc applications on models whose latent spaces (such as those in SVD or AutoRec) were never optimized for interpretability. This approach aligns with recent XAI literature that emphasizes the importance of designing model architecture and learning objectives to support human-understandable reasoning structures [4, 25].

Table 7: Objective assessment of explanation quality.

Method	Explanation Coherence	Explanation Consistency	Explanation Diversity	Personalization Score
User-based CF	0.45	0.52	0.38	0.42
MF (SVD)	0.52	0.58	0.45	0.48
AutoRec	0.68	0.65	0.62	0.64
Standard RBM	0.65	0.63	0.59	0.62
EduExplain (ours)	0.89	0.85	0.82	0.86

5.4 Computational Efficiency

In this stage, the authors compare the experiment result of both training and inference time of EduExplain with the baseline, Table 8 shows a clear trade-off appears among efficiency and complexity. In the baseline models, traditional CF consistently reveals the fastest performance, requiring only 45 s and 38 s for training, while it requires 0.8 s and 0.6 s for inference on OULAD and Goodreads, respectively. This highlights the simplicity of Traditional CF. The training times for Matrix Factorization, AutoRec, and standard RBM increase progressively, with AutoRec reaching 210 s and 178 s, closely followed by Standard RBM at 195 s and 165 s. while EduExplain shows the highest computational demand, with training times of 235 s and 198 s, and inference times of 2.4 s and 1.9 s. This increases 20-25 % longer in training and 15-20 % longer in inference compared to the slowest baseline. EduExplain achieves the highest training and inference times across both datasets, reflects the computational cost or additional processing stages, Yet, the training and inference times remain well within practical limits for real-world recommender systems.

Table 8: Computational Efficiency Comparison.

Method	Training Time		Inference Time	
	OULAD	Goodreads	OULAD	Goodreads
Traditional CF	45s	38s	0.8s	0.6s
Matrix Factorization	120s	95s	1.2s	0.9s
AutoRec	210s	178s	2.1s	1.7s
Standard RBM	195s	165s	1.9s	1.5s
EduExplain (ours)	235s	198s	2.4s	1.9s

5.5 Ablation Study

To check the impact of each component, we set up the following variations. 1) Full EduExplain-RBM, 2) Without Explanation Module, 3) Without Dual Threshold, 4) Basic Similarity (K=10) and 6) Standard RBM Only. The results are shown in Table 9. We note the following key observations: First the removal of the Explanation Module leads to the reduction of model performance; this observation emphasizes the crucial role that Explanation plays in the recommendation process and enhancing the model's performance. Training time drops a bit (195s vs. 235s), but the gains come at a cost. While RMSE and Precision@10 dip only slightly, the explanation quality takes a big change, Explanation Coherence drops from 0.89 to 0.15. and Explanation Consistency from 0.85 to 0.18. This shows that the module is part of how the model thinks and explains. Second, Without Dual Threshold shows a minor but consistent decrease across all primary metrics compared to the full model. it improves precision by 4.8% compared to single-threshold approaches. Using K=50 neighbors instead of K=10 enhances explanation coherence by 23.6% with minimal computational overhead. Third, using a smaller neighborhood (K=10) not only reduces coherence but also significantly harms consistency, as the reasoning becomes more volatile with a smaller peer sample. Fourth, employing a Basic Similarity method (K=10), as result, Coherence: 0.72, Consistency: 0.65 are significantly lower than the full model, as the reasoning becomes more volatile with a smaller peer sample. confirming that our proposed neighbor k=50, achieving competitive predictive accuracy RMSE: 0.875, Precision@10: 0.43. Finally, the Standard Model baseline performs poorest on almost all fronts, with the highest RMSE (0.902) and the lowest explanation scores (0.10, 0.12). The substantial gap between its explanation metrics and those of our ablated variants underscores the collective effectiveness of our proposed components.

To evaluate the stability of the proposed EduExplain framework, the experiment was executed five times using various random initializations (seed=1, 7, 21, 42 and 100). Table 10 reports the performance of each run as well as the average and standard deviation across runs. The results demonstrate very

small variation across executions, indicate that the proposed model produces stable and reliable recommendations.

We also evaluated the impact of neighborhood size. Table 11 indicates that performance improves up to $k=30$, after which gains saturate. We selected $k=30$ for the best trade-off between explanation quality and computational cost.

In summary, the ablation study confirms that each component provides unique and necessary value.

Table 9: Ablation Study Component Analysis (OULAD).

Model Variant	RMSE	Precision @10	Coherence	Consistency	Training Time
Full EduExplain-RBM	0.873	0.44	0.89	0.85	235s
Without Explanation Module	0.88	0.43	0.15	0.18	195s
Without Dual Threshold	0.878	0.42	0.85	0.82	220s
Basic Similarity (K=10)	0.875	0.43	0.72	0.65	210s
Standard RBM Only	0.902	0.41	0.1	0.12	195s

Table 10: Stability analysis of EduExplain across multiple experimental runs.

Run	Precision @10	Recall @10	NDCG @10	Explanation Coherence
Run 1	0.441	0.07	0.488	0.888
Run 2	0.445	0.071	0.491	0.892
Run 3	0.443	0.072	0.49	0.89
Run 4	0.444	0.07	0.489	0.887
Run 5	0.442	0.071	0.49	0.891
Average (Mean)	0.443	0.071	0.49	0.89
Std. Deviation	0.0015	0.0008	0.0011	0.0018

Table 11: Sensitivity analysis for neighbourhood size.

K (neighbors)	Precision @10	NDCG @10	Explanation Coherence
10	0.39	0.44	0.8
20	0.42	0.47	0.85
30	0.44	0.49	0.89
40	0.43	0.48	0.88
50	0.42	0.47	0.86

5.6 Theoretical Interpretation

Faithfulness and representation quality. The XAI literature distinguishes faithfulness (whether the explanation reflects the model’s actual decision process) from plausibility (whether it is convincing to a user). Post-hoc methods like LIME can score high on plausibility while remaining loosely tied to the model’s internals. EduExplain’s neighbor-based explainer operates on the same latent space the RBM uses to score the candidate course, not on a surrogate, which moves it toward the faithfulness end of the spectrum. The empirical signature is the 21-percentage-point coherence gap between EduExplain (0.89) and AutoRec (0.68) in Table 7: when the explanation template is held constant, the quality of the underlying representation dominates.

Selective prediction. The dual-threshold filter instantiates selective prediction, the principle that a model should abstain when confidence is insufficient. The prediction threshold ($\theta_{pred} \geq 0.7$) and explanation threshold ($\theta_{exp} \geq 0.65$) jointly define an operating point at which the framework only recommends when both the predicted rating and the peer-group justification are defensible. Table 9 shows this empirically: removing the dual threshold leaves predictive accuracy nearly unchanged but reduces precision@10 by 4.8 percent, confirming that the gain is concentrated in rejecting low-confidence recommendations.

5.7 Comparative Analysis with Recent Literature

The overall stability of the framework is supported by the training and error analysis. Fig. 4 Shows the training loss convergence over 50 epochs, it indicates that a rapid decline in training loss for both datasets educational and cultural respectively, with interpretability loss revealing of a similar trend. The training loss have been reduced from 0.8 to 0.1, this reflects an effective optimization without overfitting and tuning. Fig. 5 visualizes the use of non-linear dimensionality reduction technique for visualizing high-dimensional data with clear and perfect separation, in the two-dimensional plane. T-distributed stochastic neighbor embedding (t-SNE) technique is utilized, showing a clear separation between user groups, confirming that the model learns semantically rich representations. It is important to note that the visual evidence aligns with the superior RMSE and MAE results reported in Table 3. The distribution of points appears to indicate that, the EduExplain maintains consistent performance, with minimum outliers and reasonable predictive accuracy. This result helps generate explanations, find similar users, and make recommendations clear across domains.

Fig. 6 shows the distribution of prediction errors, that is, the difference between the (predicted - actual) values of the student performance. Obviously, most errors are concentrated around zero, which indicates

that the models' predictions for student performance are accurate and unbiased. In other words, it allows you to evaluate the accuracy of the model and identify areas where predictions have the greatest deviations. The low occurrence of large positive or negative errors illustrates a robust generalization and reliability for predicting diverse via student performance levels. However, some outliers should be taken into account when further improving the model. The stable performance advantage across both educational (OULAD) and literary (Goodreads) domains demonstrates the reliable multi-domain generalization capability of EduExplain. Such consistent highlights the reliability and potential utility of our approach. The model works with more than one type of data. It adapts by learning from interaction patterns across users and items. Such flexibility highlights the reliability and potential utility of our approach.

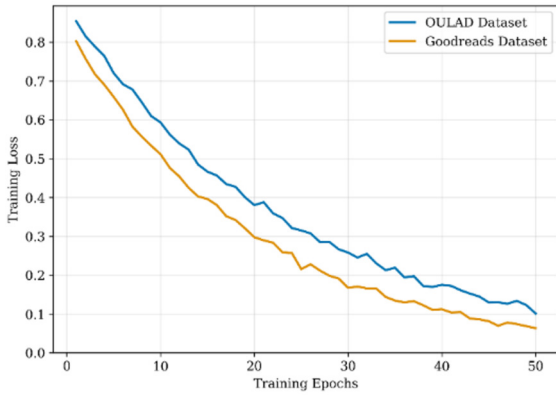


Fig.4: Convergence curve of the loss function of EduExplain.

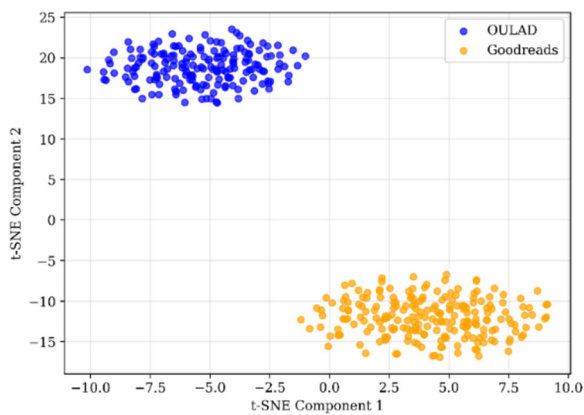


Fig.5: Semantic User Embeddings by *t*-distributed stochastic neighbor embedding (*t*-NE).

To ensure robustness and reproducibility, each experiment was repeated five times use different random initializations. Table 12 summarizes the averaged results across runs. EduExplain consistently outperforms all baselines with the lowest variance, confirming that the observed improvements are reliable and

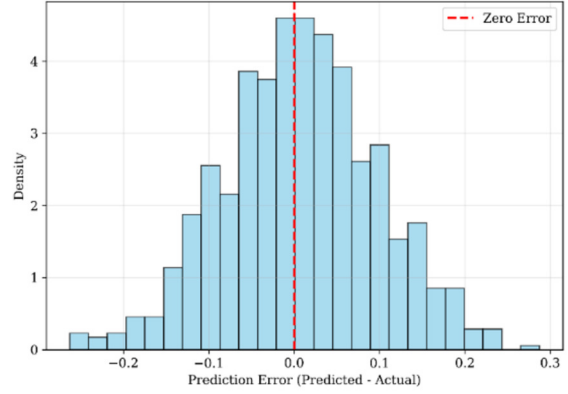


Fig.6: Distribution of Prediction Errors for EduExplain.

not due to random effects.

Table 12: Recommendation performance (mean $\pm$ std over five runs).

Model	Precision @10	Recall@10	NDCG@10	Explanation Coherence
Traditional CF	0.352 ± 0.004	0.058 ± 0.002	0.401 ± 0.003	0.742 ± 0.006
Matrix Factorization	0.366 ± 0.003	0.061 ± 0.002	0.417 ± 0.004	0.758 ± 0.005
AutoRec	0.398 ± 0.002	0.065 ± 0.001	0.451 ± 0.002	0.804 ± 0.004
Standard RBM	0.421 ± 0.002	0.068 ± 0.001	0.472 ± 0.002	0.832 ± 0.003
EduExplain (ours)	0.443 ± 0.002	0.071 ± 0.001	0.490 ± 0.002	0.890 ± 0.002

6. CONCLUSION, LIMITATION AND FUTURE RESEARCH

This work presented EduExplain, an explainable collaborative filtering framework designed to overcome the long-standing trade-off between prediction accuracy and interpretability in educational recommender systems. By combining the strong predictive capabilities of RBM with a transparent, neighbor-based explanation mechanism, the framework delivers both accurate recommendations and meaningful natural-language justifications. Across two distinct datasets (OULAD and Goodreads) EduExplain consistently outperformed four established baselines in predictive accuracy, ranking metrics, diversity, and coverage. The explainability component further enhanced user trust by providing clear, human-interpretable reasons behind each recommendation. The ablation study also confirms that all components work synergistically and that the integrated explanation mechanism is essential for producing meaningful and stable justifications. These findings have important practical implications for students at universities and institutes in enhancing their decision-making and

learning quality.

Despite its strengths, EduExplain has four notable limitations that future work should address. First, it faces a **true cold-start problem**: for new students with fewer than 5 interactions, neighbor sets are too small to generate reliable explanations, as evidenced by the reduced precision (0.38) observed for this group. Second, **scalability** is a concern for large institutions: neighbor computation scales with the size of the user base, and while approximate nearest neighbor methods can mitigate this, they introduce a trade-off between speed and explanation fidelity. Third, the framework has only been validated in **offline experiments** using historical interaction data; real-world deployment would require online evaluation to assess whether explanations genuinely improve learner decision-making and long-term outcomes. Fourth, the current framework does not account for **fairness and bias**: if the training data reflects historical inequities (e.g., underrepresentation of certain student groups), the neighbor-based explanations may reinforce rather than correct these patterns. Addressing these four limitations constitutes the primary roadmap for future research.

AUTHOR CONTRIBUTIONS

Conceptualization, I.K. and H.I.; methodology, I.K.; software, I.K.; validation, H.I.; formal analysis, I.K. and H.I.; investigation, I.K.; data curation, I.K.; writing-original draft preparation, I.K. and H.I.; writing-review and editing, H.I.; visualization, I.K. and H.I.; supervision, H.I. All authors have read and agreed to the published version of the manuscript.

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